NAME: Sumit Kumar

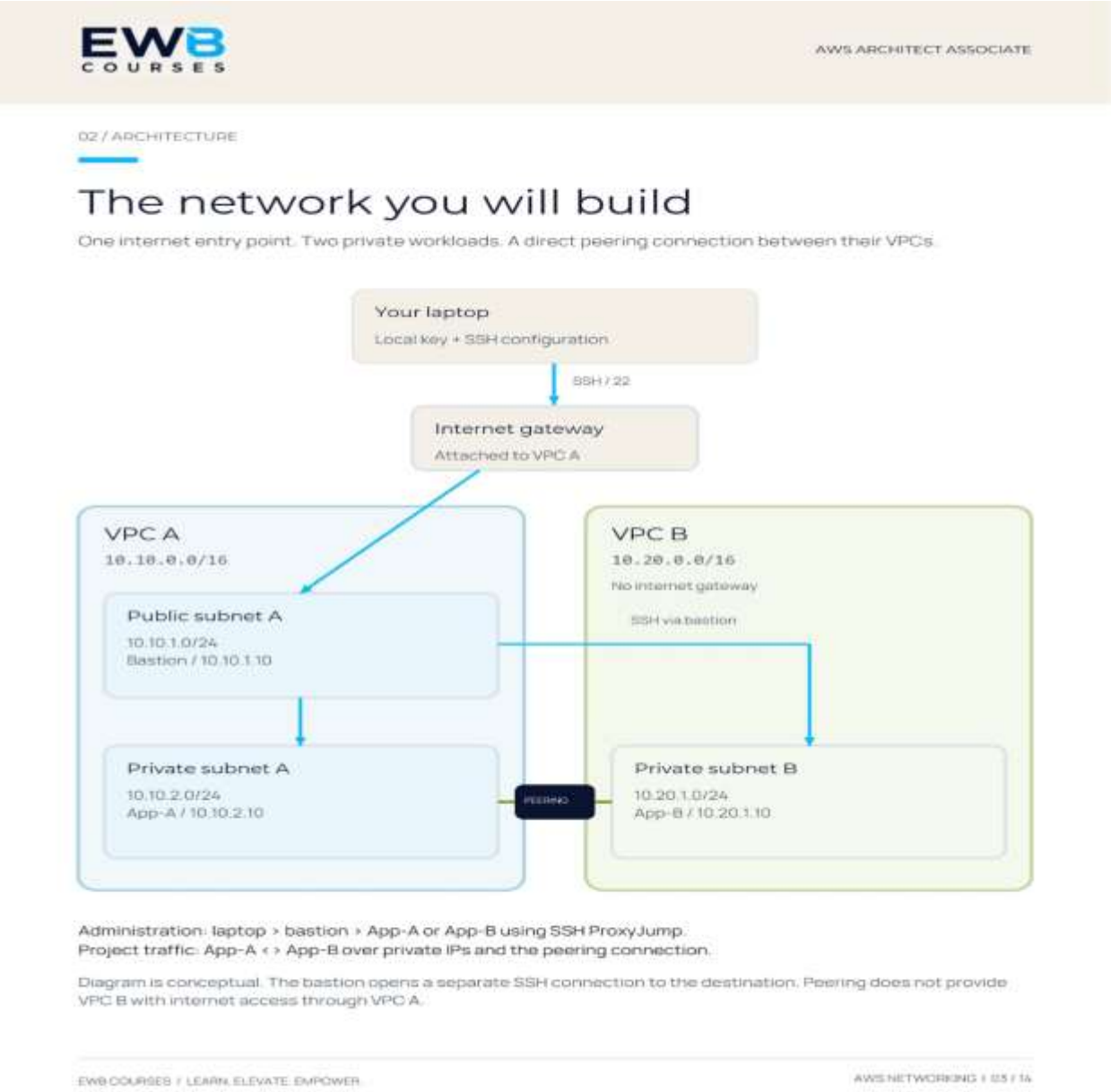
BATCH: AWS Internship Batch 1

PROJECT TITLE: AWS VPC peering Implementation

AWS REGION: ap-southeast-2(Sydney)

COMPLETION DATE: 30 Oct. 26

ARCHITECTURE



1. VPCs & Subnets Details:

VPC A Name: ewb-vpc-a | CIDR: 10.10.0.0/16

VPC B Name: ewb-vpc-b | CIDR: 10.20.0.0/16

Subnets in VPC A:

Public Subnet: ewb-a-public | CIDR: 10.10.1.0/24

Private Subnet: ewb-a-private | CIDR: 10.10.2.0/24

Subnets in VPC B:

Private Subnet: ewb-b-private| CIDR: 10.20.1.0/24

3. Route Tables & Associations

ewb-rt-a-public: Associated with subnet-06b002a08c376c174 / ewb-a-public

ewb-rt-a-private: Associated with subnet-02b96ef513d063edd/ ewb-a-private

ewb-rt-b-private: Associated with subnet-020a6436e2bb11fdc / ewb-b-private

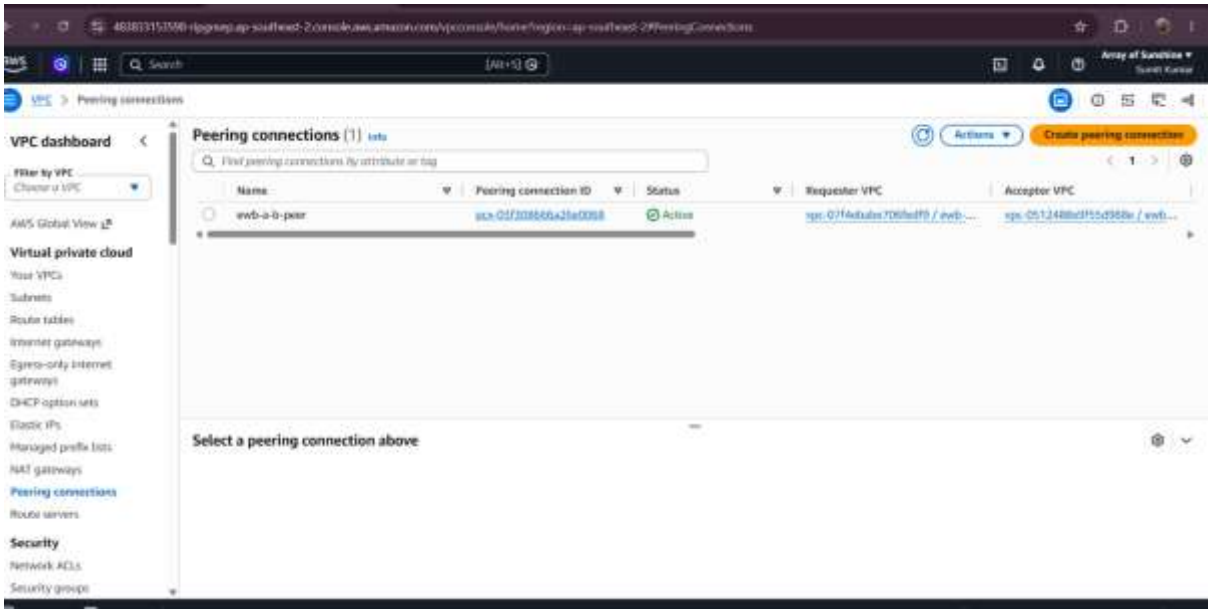
4. EC2 Instances Details

Bastion Host: Instance ID = i-0a8bab5a4eee1d6b2 | Private IP = 10.10.1.10

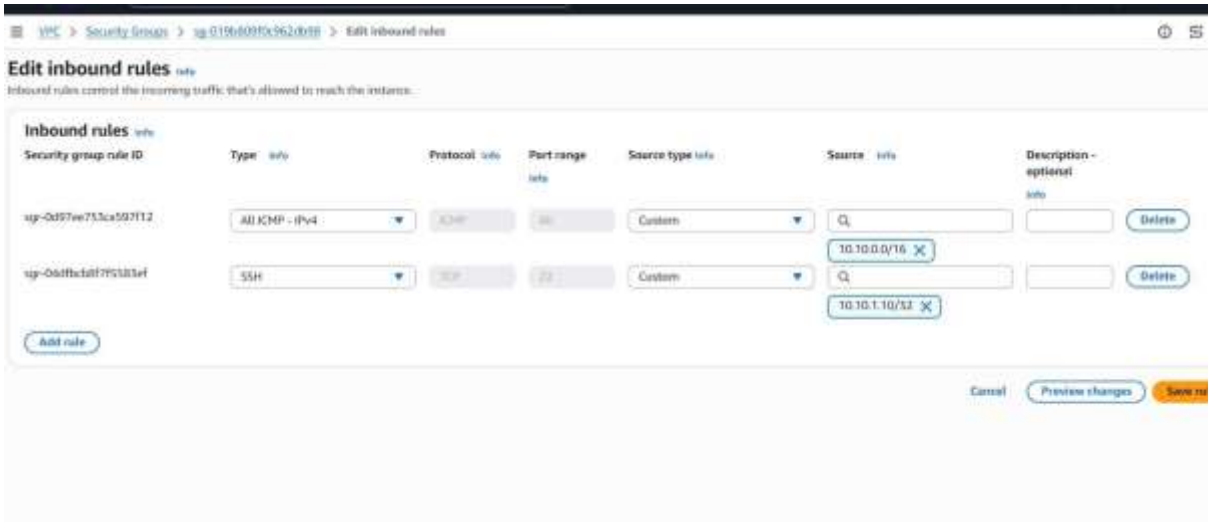
App-A: Instance ID = i-0d7e29f8affc28662 | Private IP = 10.10.2.10

App-B: Instance ID = i-0fefc905cabdacea3 | Private IP = 10.20.1.10

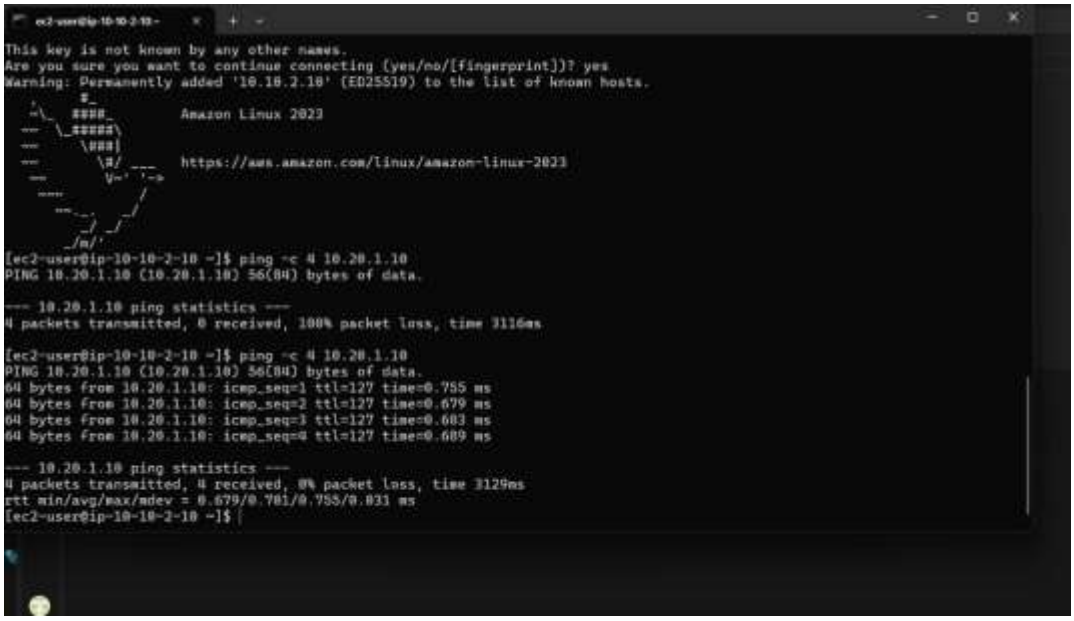
ACTIVE VPC Peering:

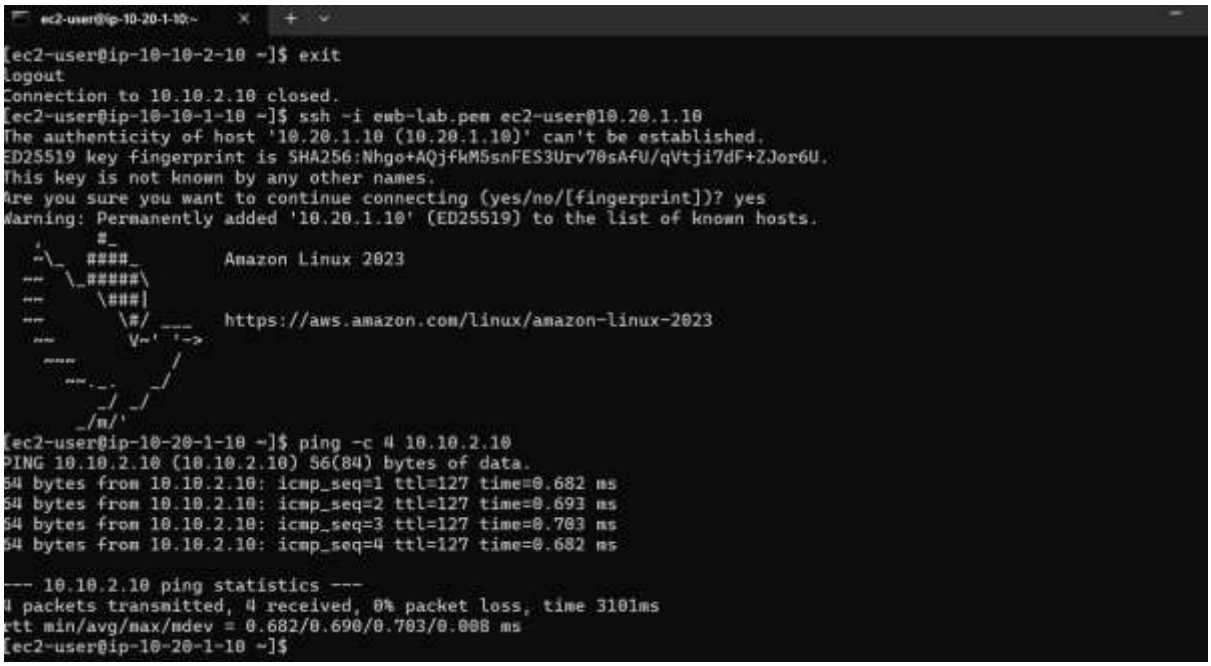


Security Groups:

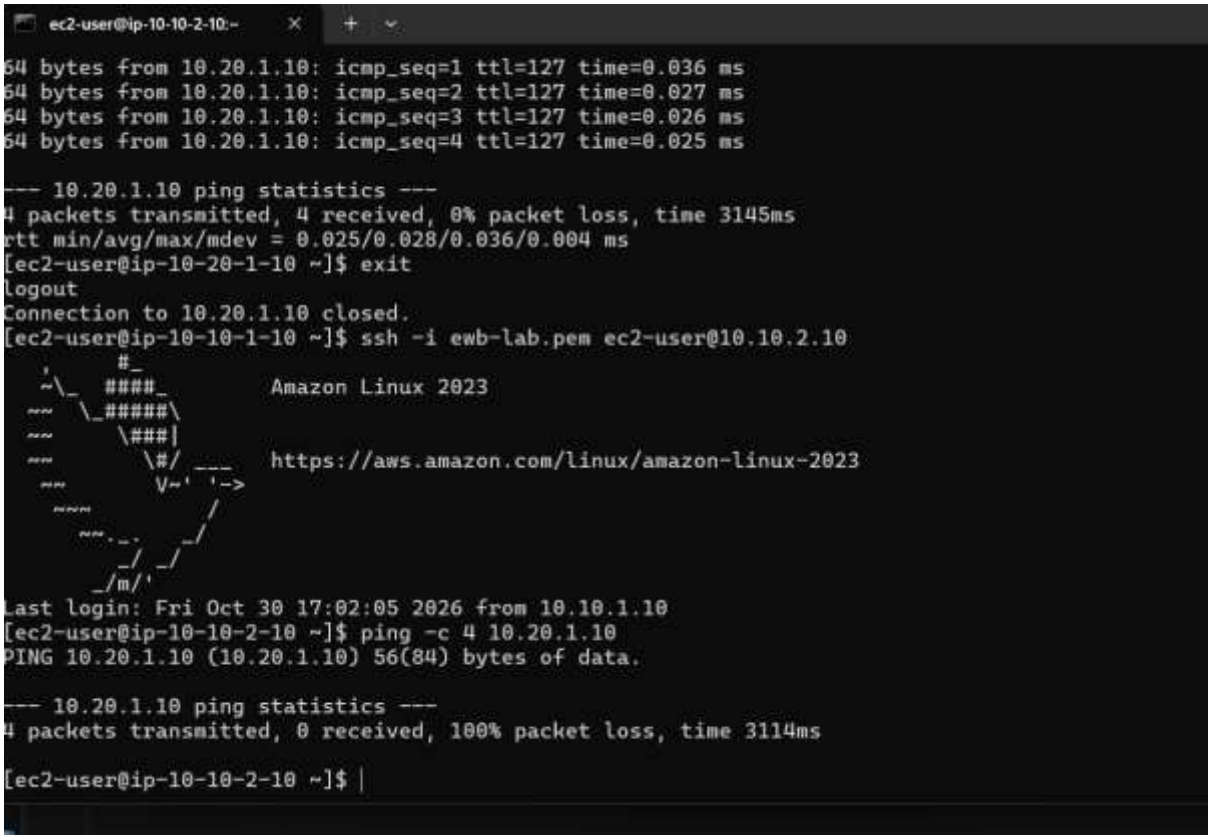


Test 2 & 3:

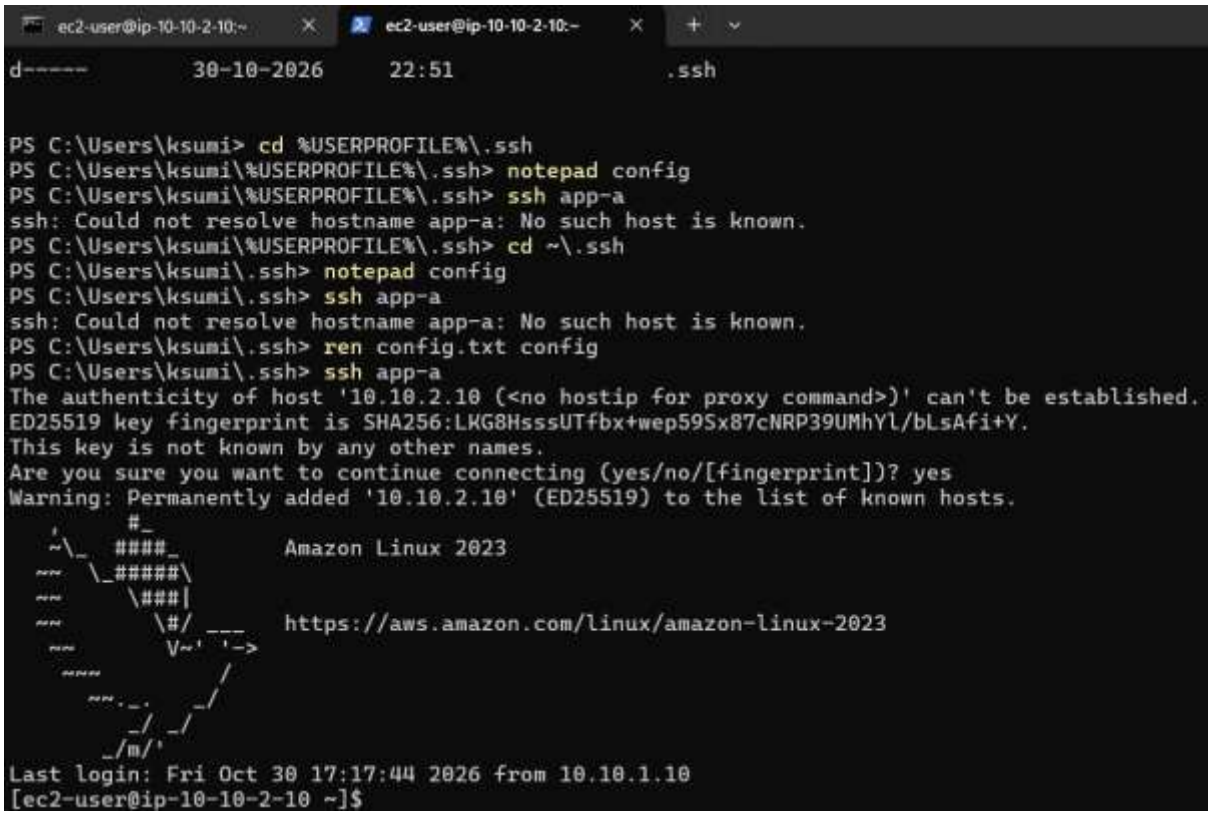

This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3116ms
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.755 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.679 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.683 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.689 ms
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3129ms
rtt min/avg/max/mdev = 0.679/0.701/0.755/0.031 ms
[ec2-user@ip-10-10-2-10 ~]$


[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
[ec2-user@ip-10-10-1-10 ~]$ ssh -i ewb-lab.pem ec2-user@10.20.1.10
The authenticity of host '10.20.1.10 (10.20.1.10)' can't be established.
ED25519 key fingerprint is SHA256:Nhgo+AQjfkM5snFES3Urv70sAfU/qVtji7dF+ZJor6U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.682 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.693 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.703 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.682 ms
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3101ms
rtt min/avg/max/mdev = 0.682/0.690/0.703/0.008 ms
[ec2-user@ip-10-20-1-10 ~]$

Test 4:

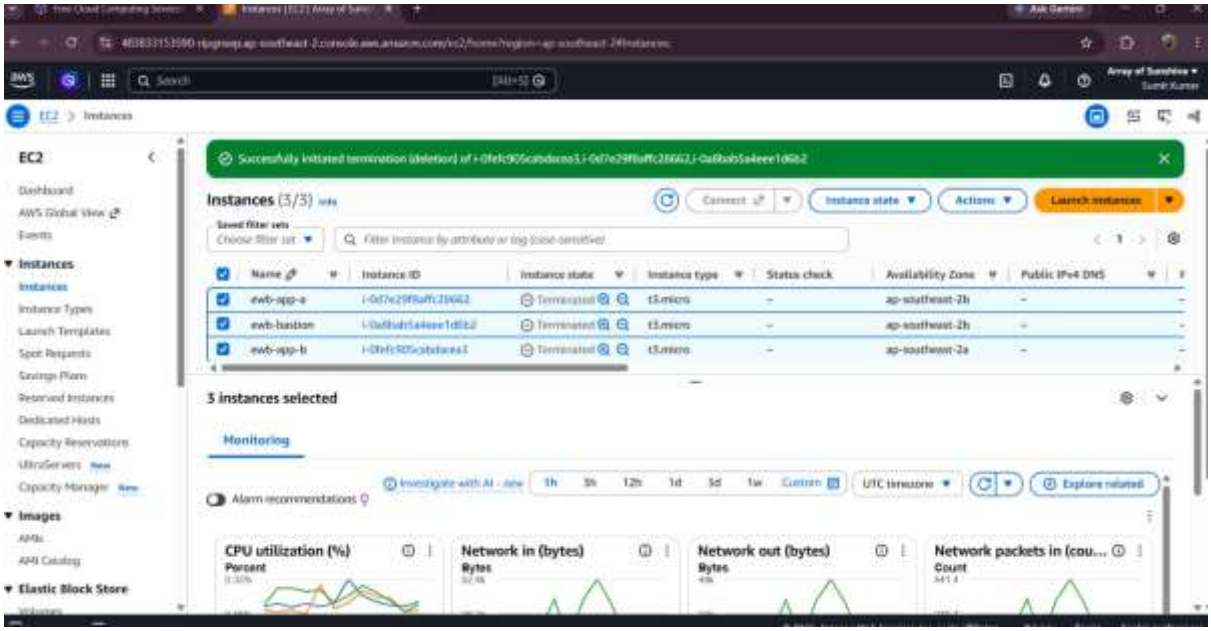

ec2-user@ip-10-10-2-10:~
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.036 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.027 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.026 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.025 ms
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3145ms
rtt min/avg/max/mdev = 0.025/0.028/0.036/0.004 ms
[ec2-user@ip-10-20-1-10 ~]$ exit
logout
Connection to 10.20.1.10 closed.
[ec2-user@ip-10-10-1-10 ~]$ ssh -i ewb-lab.pem ec2-user@10.10.2.10
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Fri Oct 30 17:02:05 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3114ms
[ec2-user@ip-10-10-2-10 ~]$

Test 5:


ec2-user@ip-10-10-2-10:~
ec2-user@ip-10-10-2-10:~
d------ 30-10-2026 22:51 .ssh
PS C:\Users\ksumi> cd %USERPROFILE%\.ssh
PS C:\Users\ksumi\%USERPROFILE%\.ssh> notepad config
PS C:\Users\ksumi\%USERPROFILE%\.ssh> ssh app-a
ssh: Could not resolve hostname app-a: No such host is known.
PS C:\Users\ksumi\%USERPROFILE%\.ssh> cd ~\.ssh
PS C:\Users\ksumi\.ssh> notepad config
PS C:\Users\ksumi\.ssh> ssh app-a
ssh: Could not resolve hostname app-a: No such host is known.
PS C:\Users\ksumi\.ssh> ren config.txt config
PS C:\Users\ksumi\.ssh> ssh app-a
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:LKG8HsssUTfbx+wep59Sx87cNRP39UMhYl/bLsAfi+Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Fri Oct 30 17:17:44 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$

Cleanup Proof:

EC2 deletion:



My learnings:

"Through this project, I learned how to set up a VPC peering connection between two different VPCs. I configured public and private route tables, accessed private instances via SSH using a Bastion host, and practically tested network traffic control by modifying ICMP rules in Security Groups."